

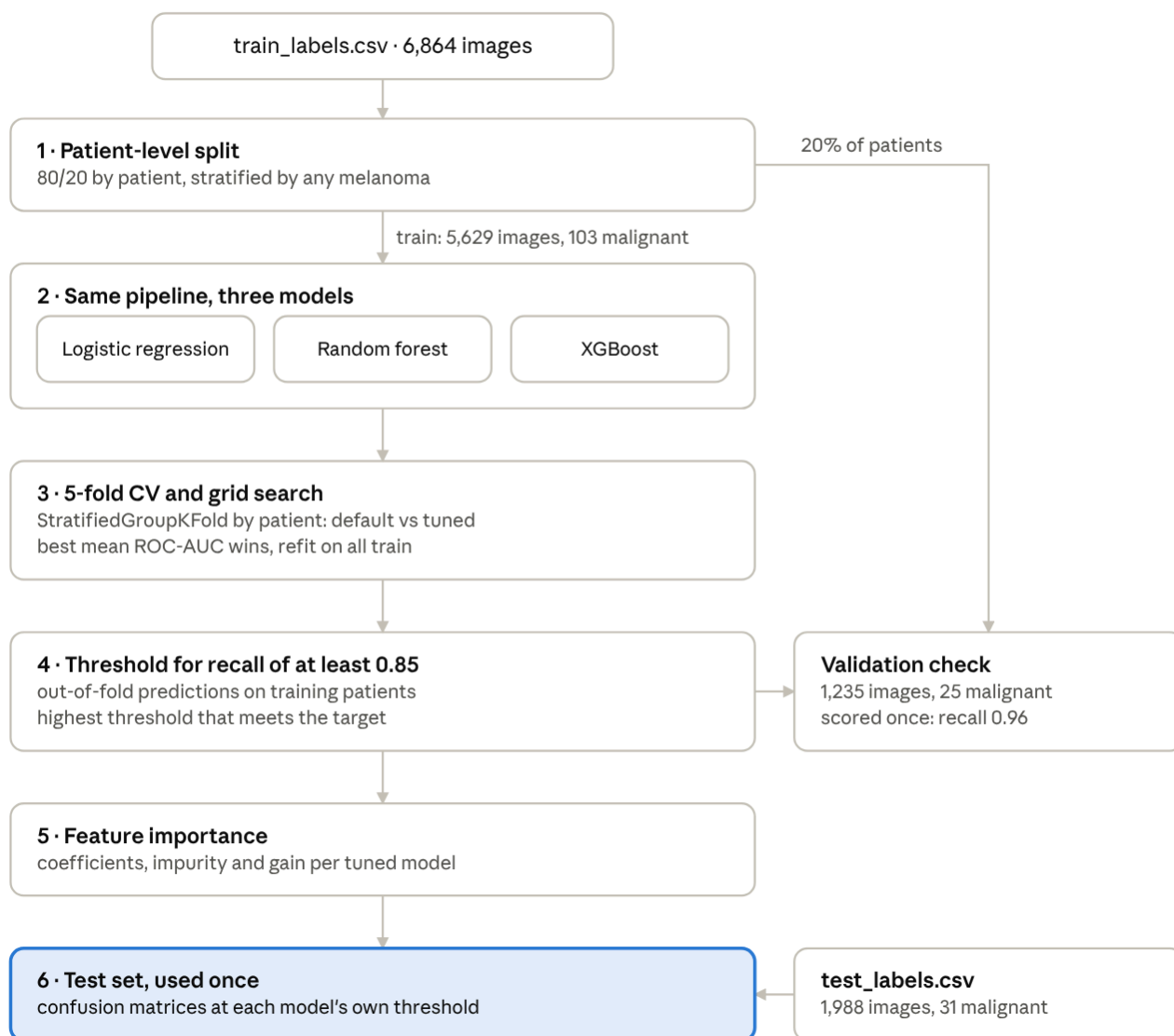
Metadata Melanoma Model: Modelling Pipeline Report

Oct 8, 2026 · @Joshiah

Pipeline overview

The metadata model (age, sex and anatomical site) was built in six steps, and logistic regression came out best: on the test set it caught 87% of melanomas (27 of 31) with ROC-AUC 0.679. Every modelling choice uses training patients only; validation is scored once as a check, and the test set once at the end.

Every modelling choice uses training patients; the test set is used once



Training patients flow down the left through every decision, while validation and test data enter from the right only to be scored.

Step 1: Data split

The 6,864 training images were split 80/20 by patient into train and validation.

`test_labels.csv` was held back untouched until the final step.

Set	Source	Images	Malignant	Malignant rate	Used for
Train	<code>train_labels.csv</code> , 80% of patients	5,629	103	1.83%	CV, tuning, threshold, final fit
Validation	<code>train_labels.csv</code> , 20% of patients	1,235	25	2.02%	One check of the tuned models
Test	<code>test_labels.csv</code> (106 patients)	1,988	31	1.56%	Final one-off evaluation

- **Patient-level split.** All of a patient's lesions land in the same set, so no patient is seen in both training and evaluation. The test set shares 0 patients with the training file.
- **Stratified by patient outcome.** Each patient is labelled by whether they have any melanoma, and the split keeps that proportion. An earlier unstratified split gave 1.26% vs 2.06% malignant between validation and train.
- **Cleaning.** Sex is encoded 0/1, missing anatomical site becomes its own `missing` category, and `diagnosis` and `benign_malignant` are dropped because they reveal the target.
- **Features.** Age, sex and anatomical site. Site one-hot encoding (and age scaling for logistic regression) sits inside each model's pipeline, so it is refit on training data only.
- **Metrics.** ROC-AUC and PR-AUC rather than accuracy: predicting "benign" for everything would score about 98% accuracy. A no-skill model has ROC-AUC 0.5 and PR-AUC equal to the malignant rate.

Step 2: Three candidate models

One linear model and two tree ensembles were compared through identical pipelines, so differences come from the model alone. Trees are included because malignancy risk is flat (about 1 to 1.5%) below age 60 and then jumps to 4.6 to 6.6%, a threshold shape that splits capture directly and a single linear age coefficient cannot.

Model	Base settings	Class imbalance handling	Preprocessing in pipeline
Logistic regression	<code>liblinear</code> solver, <code>max_iter=1000</code>	<code>class_weight='balanced'</code>	One-hot site, standardised age
Random forest	200 trees	<code>class_weight='balanced'</code>	One-hot site
XGBoost	200 boosting rounds, log-loss	<code>scale_pos_weight =</code> benign/malignant ratio (about 54)	One-hot site

With only about 1.8% malignant images, every model reweights the classes so a malignant lesion counts roughly 54 times as much as a benign one during training.

Step 3: 5-fold CV and hyperparameter tuning

Tuning lifted every model's mean CV ROC-AUC, most for random forest (0.580 to 0.620), but all gains are smaller than the fold-to-fold spread. Logistic regression stays best on both metrics.

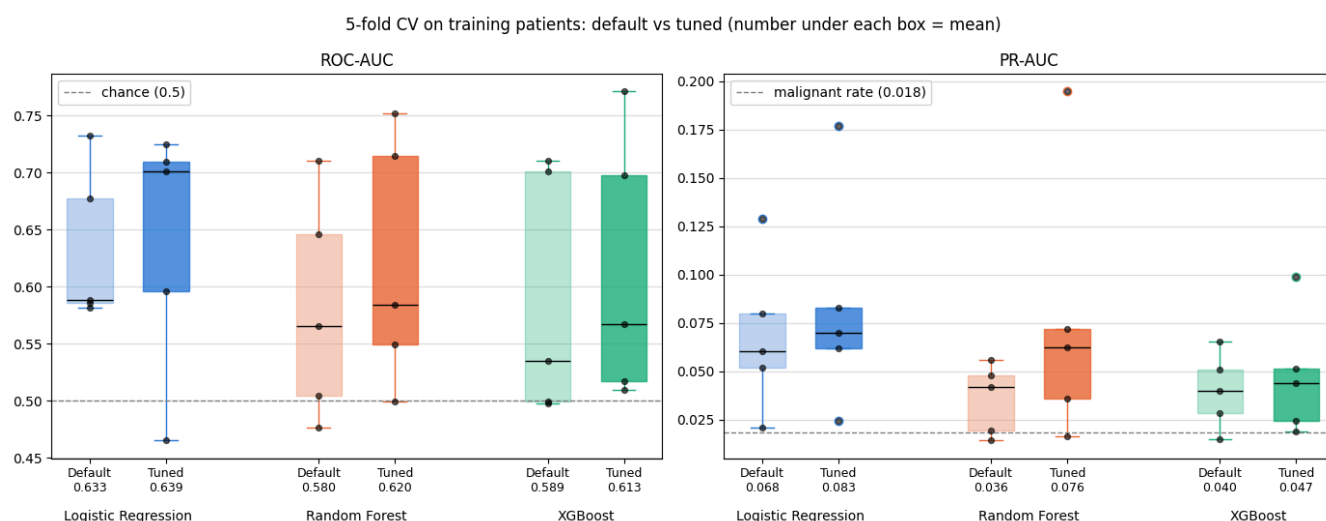
How it was run (all on the training patients; validation untouched):

- Folds.** `StratifiedGroupKFold` with 5 folds, grouped by patient and stratified by target, so no patient spans a fold's training and held-out parts. Each fold holds about 20 malignant images.
- Default baseline.** Each model's pipeline with its base settings is scored on the 5 held-out folds.
- Grid search.** Every grid setting is scored on the same 5 folds. The setting with the highest mean ROC-AUC wins (`refit='ROC-AUC'`); PR-AUC is recorded but not used to choose.
- Refit.** The winning setting is retrained on all training patients and becomes the model's final version.

Model	Grid size	Selected setting	Default ROC-AUC	Tuned ROC-AUC	Default PR-AUC	Tuned PR-AUC
Logistic regression	12	<code>C=0.001</code> , L2	0.633	0.639	0.068	0.083
Random forest	40	<code>max_depth=3</code> , <code>max_features='sqrt'</code> , <code>min_samples_leaf=100</code>	0.580	0.620	0.036	0.076

Model	Grid size	Selected setting	Default ROC-AUC	Tuned ROC-AUC	Default PR-AUC	Tuned PR-AUC
XGBoost	54	max_depth=2, learning_rate=0.05, min_child_weight=10, subsample=1.0	0.589	0.613	0.040	0.047

The no-skill PR-AUC is 0.018 (the training malignant rate).



- **Every model chose heavy regularisation.** LR picked the smallest `C` (strongest penalty), and both tree models picked shallow trees with large minimum leaf sizes. With three features and about 100 positives, simpler models generalise better.
- **Spread is wide.** Fold standard deviations are 0.06 to 0.11 for ROC-AUC, and one LR fold dips below 0.5. Differences between models of a few hundredths are within noise.
- **Tuned scores are slightly optimistic.** Each is the best of 12 to 54 settings scored on these same folds, so the validation and test results are the fairer comparison.

Step 4: Threshold selection (recall of at least 0.85)

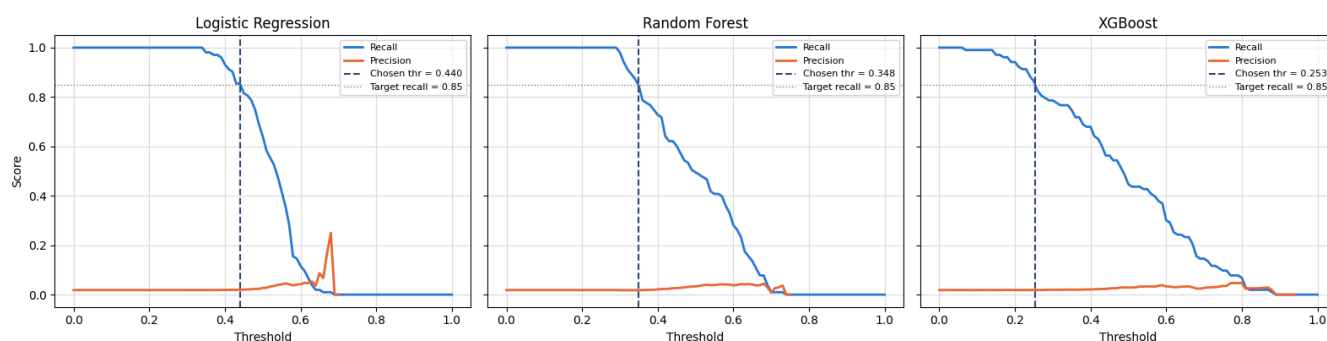
To catch at least 85% of melanomas, all three models must flag 78 to 86% of training images, with precision barely above the 1.8% base rate. Logistic regression meets the target while flagging the fewest images.

How it was picked. Each tuned model scores every training image with a model that never saw that patient (out-of-fold predictions from the same 5 folds). Scoring training rows with the final model instead would overstate recall. The threshold is the highest one that still meets the recall target, which keeps as much precision as possible:

$$t^{\{*\}} = \max \{t : \text{recall}(t) \geq 0.85\}$$

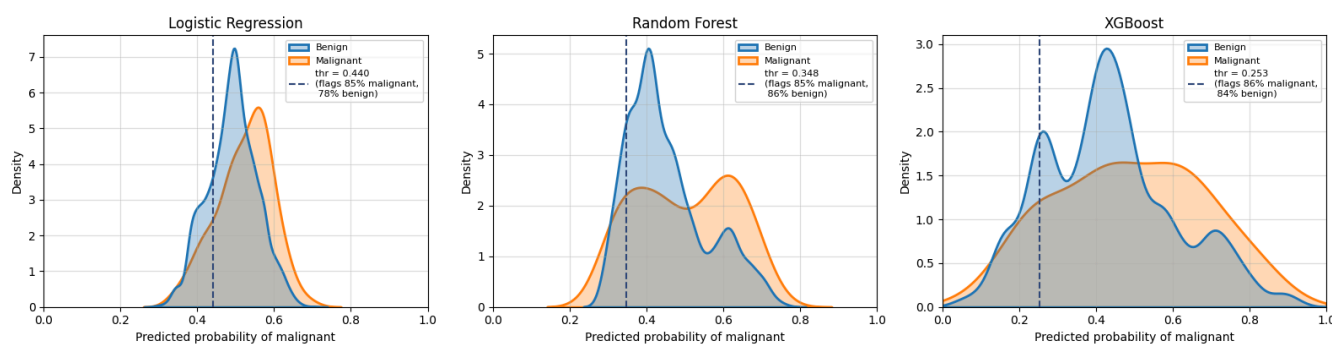
Model	OOF ROC-AUC	OOF PR-AUC	Threshold	OOF recall	OOF precision	Images flagged
Logistic regression	0.638	0.038	0.440	0.854	2.01%	77.8%
Random forest	0.606	0.031	0.348	0.854	1.82%	85.8%
XGBoost	0.586	0.027	0.253	0.864	1.87%	84.5%

Precision/Recall vs Threshold (tuned models, out-of-fold training)



Recall stays at 1.0 up to about 0.35 for LR and 0.3 for RF, then falls steeply. Precision stays near zero until thresholds where recall has collapsed, so there is no threshold with both high recall and useful precision.

Predicted probability by true class (tuned models, out-of-fold training)



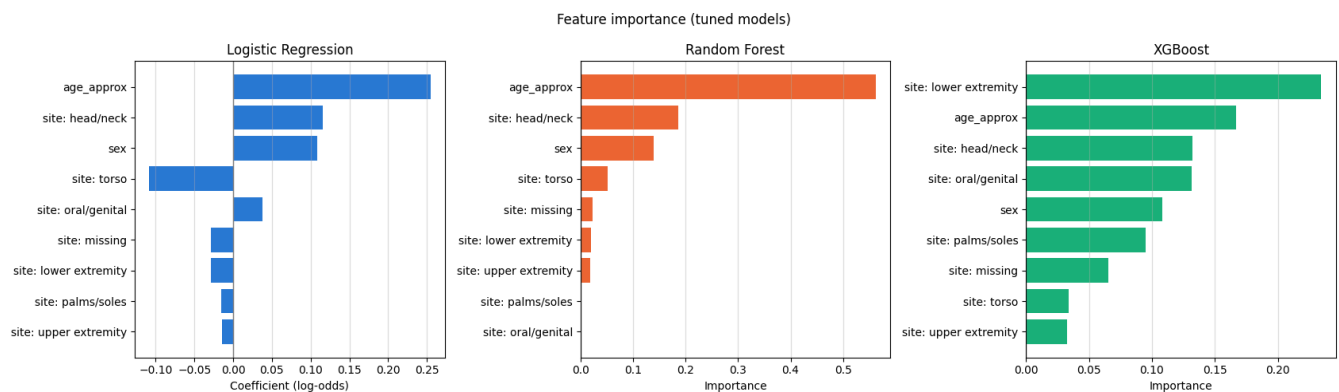
The benign and malignant distributions overlap almost completely. Random forest shows the clearest second malignant peak near 0.6, but benign images also cluster there.

Validation check. Each final model was scored once on the 1,235 validation images at its own threshold. All three caught 24 of 25 melanomas (recall 0.96).

Model	ROC-AUC	PR-AUC	Recall	Precision	Images flagged
Logistic regression	0.750	0.064	0.96	2.57%	75.7%
Random forest	0.704	0.070	0.96	2.04%	95.5%
XGBoost	0.662	0.053	0.96	2.10%	92.6%

Step 5: Feature importance

Age, head/neck site and sex rank near the top for all three tuned models; age leads for logistic regression and random forest. The scales differ by model, so compare rankings within each panel, not bar lengths across panels.



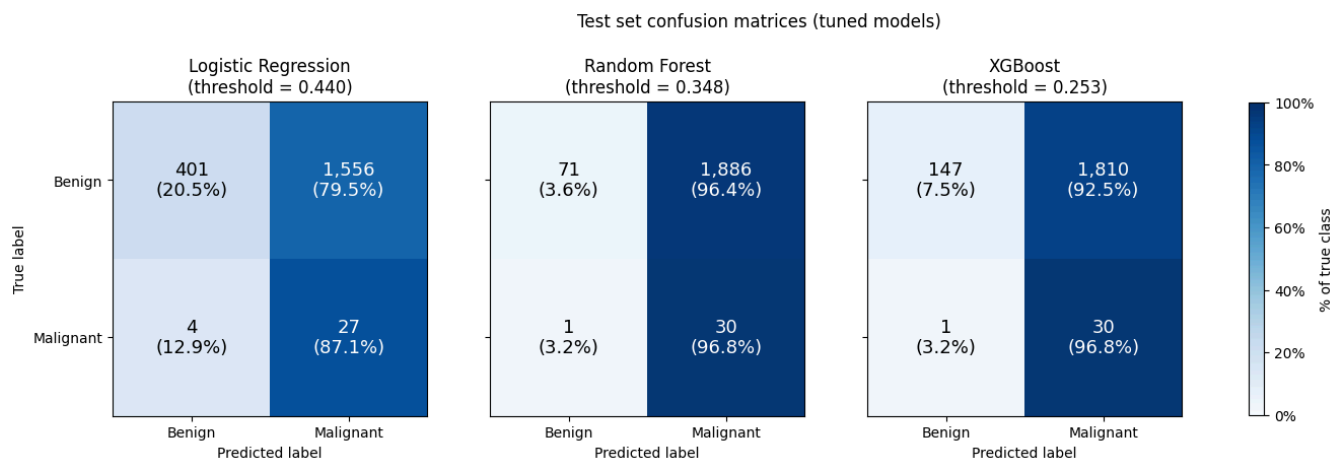
Model	Measure	Top features	Notes
Logistic regression	Coefficient (log-odds; age per 1 SD)	Age +0.25, head/neck +0.12, male +0.11, torso -0.11	Sign shows direction. Strong regularisation (<code>C=0.001</code>) shrinks every coefficient toward zero.
Random forest	Impurity decrease	Age 0.56, head/neck 0.19, sex 0.14	Palms/soles and oral/genital score 0: with <code>min_samples_leaf=100</code> , a split cannot isolate a site that has fewer than 100 training images.
XGBoost	Gain (average loss reduction per split)	Lower extremity 0.23, age 0.17, head/neck 0.13, oral/genital 0.13	Gain rewards rare categories that make one sharp split, so small sites rank higher than in the other models.

Only logistic regression shows direction: older age, head/neck lesions and male sex raise the predicted risk, and torso lesions lower it. The tree importances say how much a feature is used, not which way it pushes.

Step 6: Test set results

On the 1,988 unseen test images, logistic regression gives the most usable trade-off: it catches 27 of 31 melanomas (87.1%) while clearing 401 benign images. Random forest and XGBoost catch 30 of 31 but flag 93 to 96% of all images, close to flagging everyone.

Each model is its tuned version, trained on all training patients, applied once with its own threshold from step 4.



Model	Threshold	ROC-AUC	PR-AUC	Recall	Precision	Images flagged	TP	FN	FP	TN
Logistic regression	0.440	0.679	0.038	0.871	1.71%	79.6%	27	4	1,556	401
Random forest	0.348	0.685	0.034	0.968	1.57%	96.4%	30	1	1,886	71
XGBoost	0.253	0.661	0.033	0.968	1.63%	92.6%	30	1	1,810	147

The test malignant rate is 1.56%, so that is the no-skill PR-AUC.

- **Ranking ability is real but modest.** ROC-AUC of 0.66 to 0.69 and PR-AUC about twice the base rate show the metadata carries some signal.
- **The recall target holds.** All three met or exceeded 0.85 on new patients, so the thresholds transferred.
- **Precision does not improve.** At these thresholds precision (1.6 to 1.7%) is almost identical to the base rate. Flagged images are barely more likely to be melanoma than a random image.
- **Tree thresholds drifted.** Random forest flagged 85.8% of training images out-of-fold but 96.4% of test images, so its scores shifted between populations more than logistic regression's (77.8% to 79.6%).

Takeaways and limitations

Logistic regression is the metadata model to carry forward: it matches the tree models on test ROC-AUC, flags the fewest images at the recall target, and its scores were the most stable from training to test. Age, sex and site alone cannot separate melanoma from

benign lesions at high recall, so the metadata model is best used as an input alongside the image model rather than on its own.

- **Small positive counts.** Validation has 25 melanomas and test has 31, so one missed melanoma moves test recall by 3.2 points. Recall differences between models on test are 1 to 4 cases.
- **Tuning optimised ROC-AUC.** Choosing settings by PR-AUC, the metric that focuses on finding the rare positives, might pick different settings.
- **Validation was used only as a check.** Final models were trained on the training patients alone, not refit on train plus validation.